Workshop
FUSE + Synligare

2015-05-22

Needs identification – FUSE

FUSE
The Problem – Project Challenge

•Today, most common accident cause :
Driver misjudgments

Tomorrow:
Autonomous driving

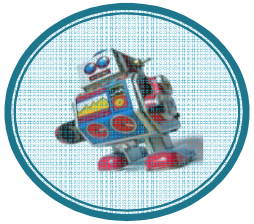
But

We need to make sure that autonomous driving is safe
(today we don't)

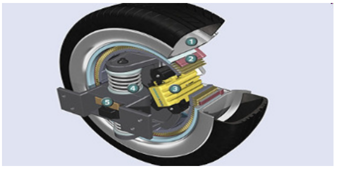
Three Dimensions of Autonomy

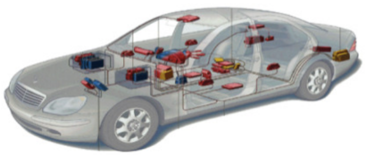
How much autonomy?

Driver only	Assisted Driver	Semi-automated	Highly automated	Fully automated



Autonomy in how much?





The ITS environment (source: ETSI)

Autonomy of how much?

City Safety	ACC, CMbB	Autopilot
One function	Set of related functions	All functions (no driver)

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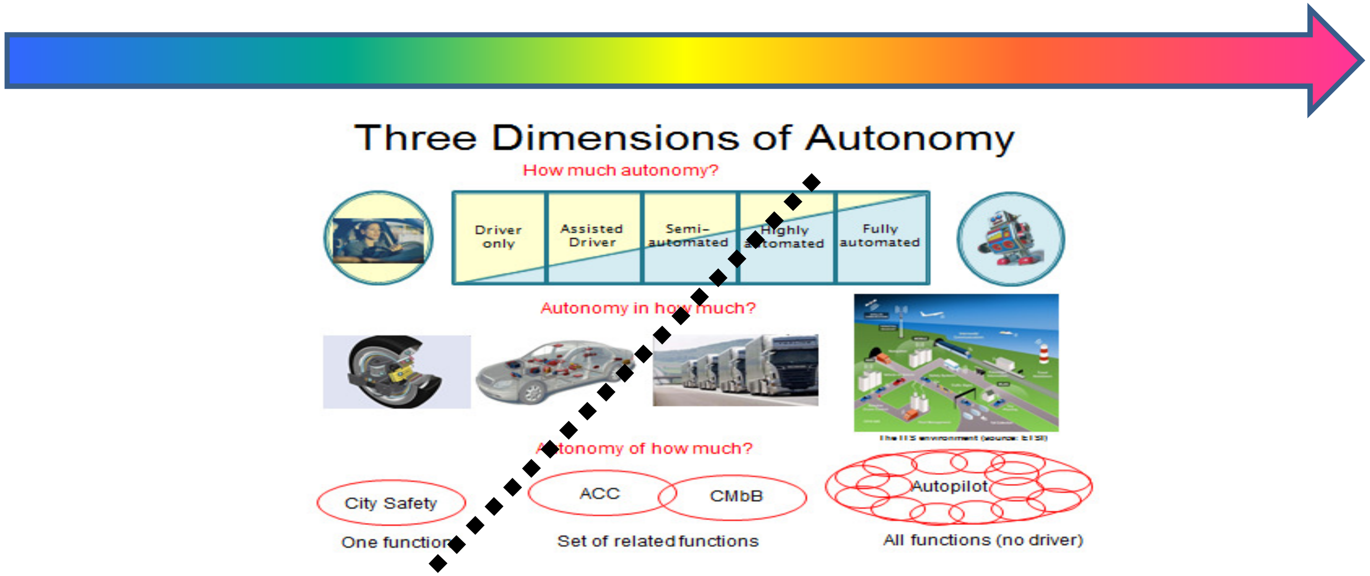
Towards More Autonomy in More



What happens to <u>Functional Safety</u> when passing the dotted line?

- How to define it? (Lacking definitions in ISO 26262)
- How to achieve it? (Demand for architectural patterns, and division of responsibility)
- How to prove it? (Demand for new compositional safety arguing)

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ISO26262 - Automated Driving and Autonomous Vehicles

- Two types of reasons why ISO26262 becomes problematic
  - Things are (much) more complicated
    - Item definition for extremely complex functionalities
    - FSC and TSC much more complex
  - Things are fundamentally different
    - Manual driver not in the loop

Focus of FUSE

- Functional safety
- Scalable Architectures
- New methods for development and safety analysis

} for Autonomy

Well complements other efforts focusing on: sensing, estimation, control strategies and algorithms, HMI, …

Drive Me

- Volvo Cars together with authorities etc.
- Self-driving cars for sustainable mobility
- Autonomous drive in commuting situations
- Project started in 2014
- On the roads 2017
  - 100 cars driven by private persons
  - 50 kilometres of commuting roads in and around Gothenburg

https://www.media.volvocars.com/global/en-gb/media/pressreleases/145619/volvo-car-groups-first-self-driving-autopilot-cars-test-on-public-roads-around-gothenburg

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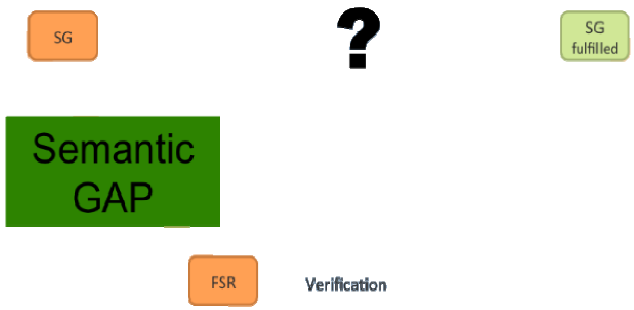
Drive Me – Some Choices and Lessons Learned

- Involvement of authorities important
  - Joint gaining of acceptance of autonomous vehicles
- Put fully autonomous cars on the market in steps
  - Focus first on use cases with high customer value
- Auto Pilot fully responsible, once activated
  - Cannot rely on driver to take over
  - Safe strategy needed for every possible scenario
- Safety Goals – Functional and Technical Safety Concepts
  - New solutions needed (work in progress)

Topics: The semantic gap

- Abstract Safety Goals for autonomous functions, e.g. ”Vehicle shall stay on road” implies that stating the connections between levels may be challenging
- => The semantic gap

- This leads to challenge of..
  - ..defining concrete and detailed FSC from SG i.e. synthesis, and..
  - ..analysing the FSC to show fulfilment of SG.
- Also corresponding ”right leg” activities verification and aggregation of safety evidence



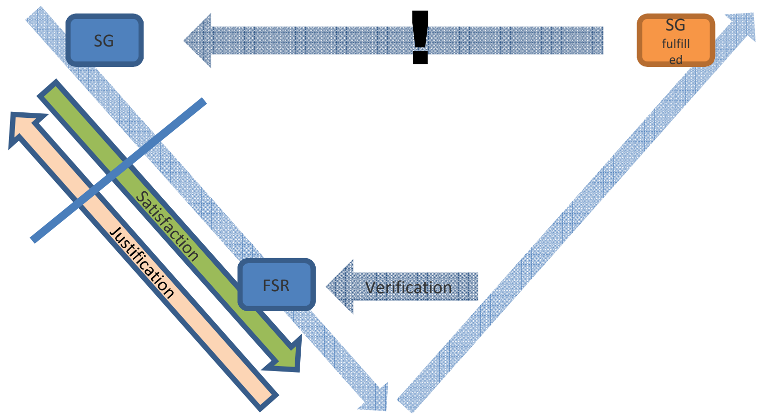
- SG <-> FSR as example, but valid for at any level, e.g. TSR <-> TSR

Topics:
Satisfaction and Justification

- Rationale for relationships between levels.
  - This is part of the safety case
  - Due to Semantic Gap the ”distance to be argued” increases
- Satisfaction - Synthesis
  - Argues what FSRs are needed to fulfil SG.
  - This is related to sound Requirements Engineering and Traceability
- Justification - Analysis
  - Argues why (and perhaps to what extent) a composition of FSRs fulfil SG.
  - This is safety related

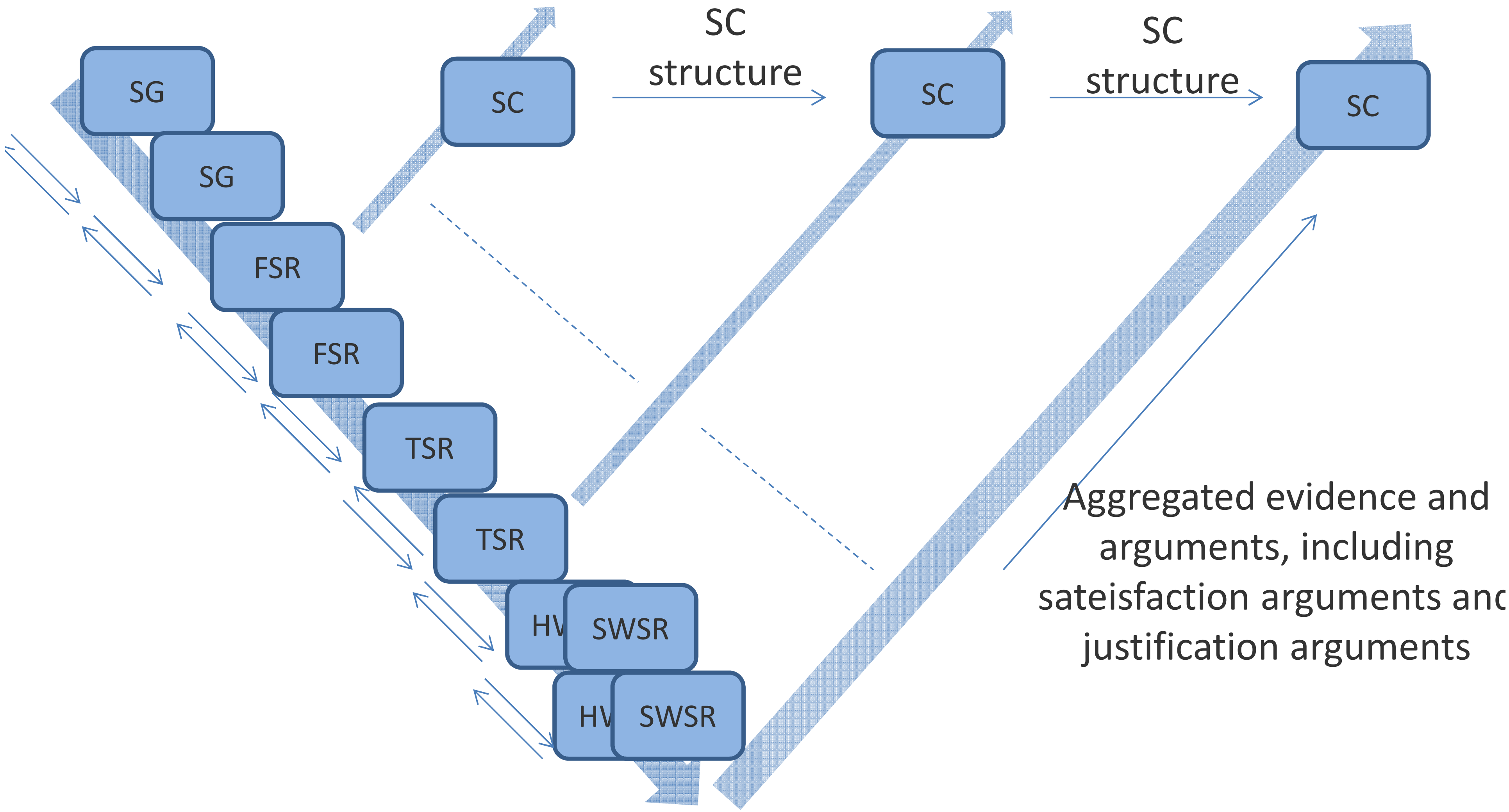
Safety requirement break-down: Concept handover

- The need for being able to make clear interfaces between stakeholders; anywhere in the lifecycle – The ”Cut”
- It shall be possible to show satisfiability and justification.



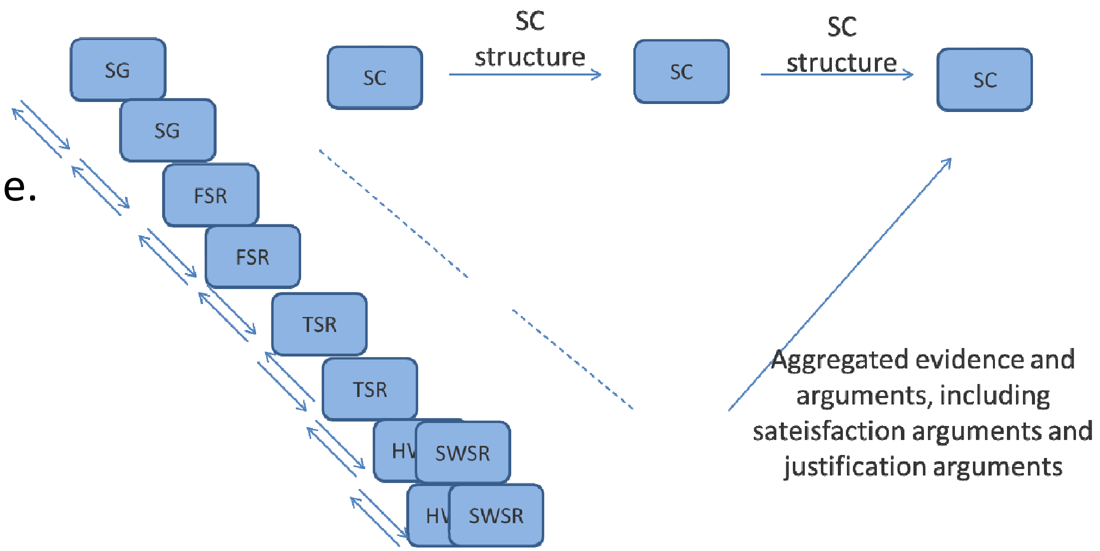
- Two development cases:
  - In context: Safety Requirements, (passed down)
  - Out of context: Safety Contract (passed up)
- What is the information that is passed at the ”Cut”?
  - Note that a DIA is not sufficient. A DIA only describes where the ”cut” takes place, not the information
- What is the meta model to structure requirements with sat / jus arguments?
- In both cases sat/jus - arguments shall be compiled into the corresponding Safety Case.

Add intra-abstraction level activities



Potential solution to Semantic Gap

- Support from reference lifecycle to make refinement within same abstraction level



- Why? Analysis!
  - To make every step manageable.
  - This is related to simplifying justification and satisfaction argumentation for rationale of requirement hierarchy.
- Why? Communication!
  - Fitting the customer/supplier relation, e.g. both have FSC or TSC
  - The customer and supplier TSC may differ
  - Customer TSC can be more detailed or reduced scope